Machine Learning Assignment 2

Student Depression Classification Model Comparison

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1. GitHub Repository

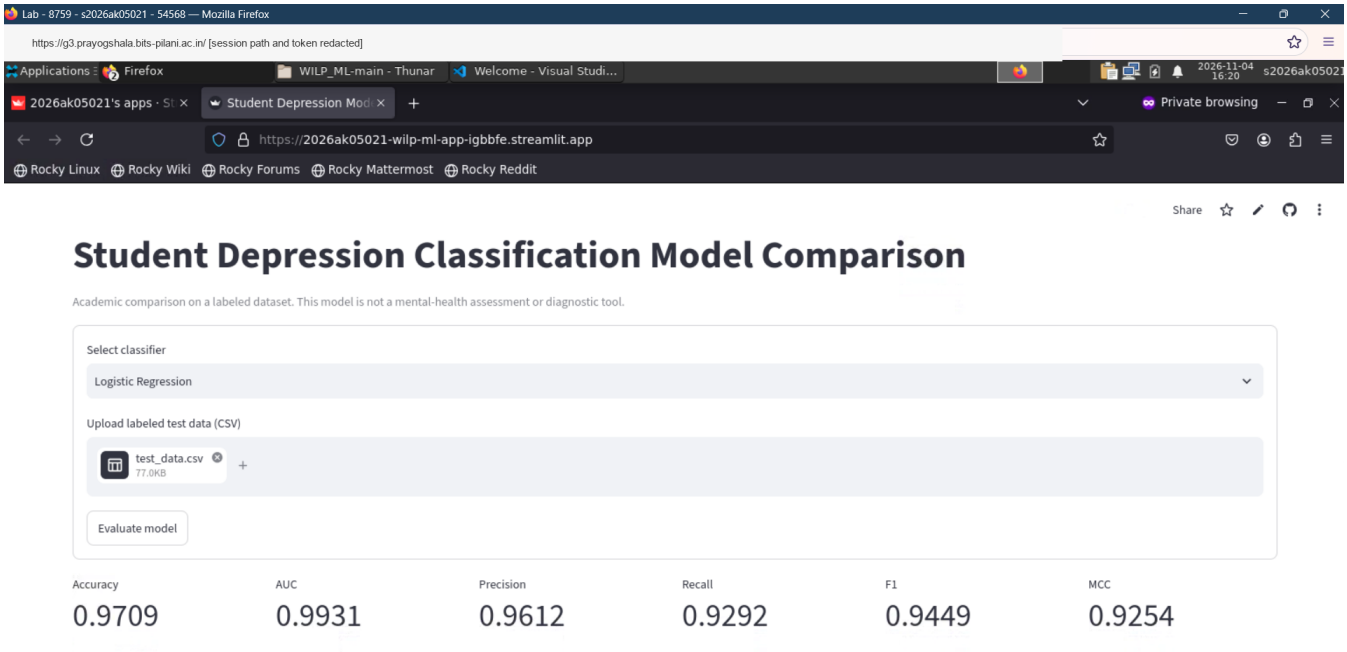
https://github.com/2026ak05021/WILP_ML/tree/master

2. Live Streamlit Community Cloud App

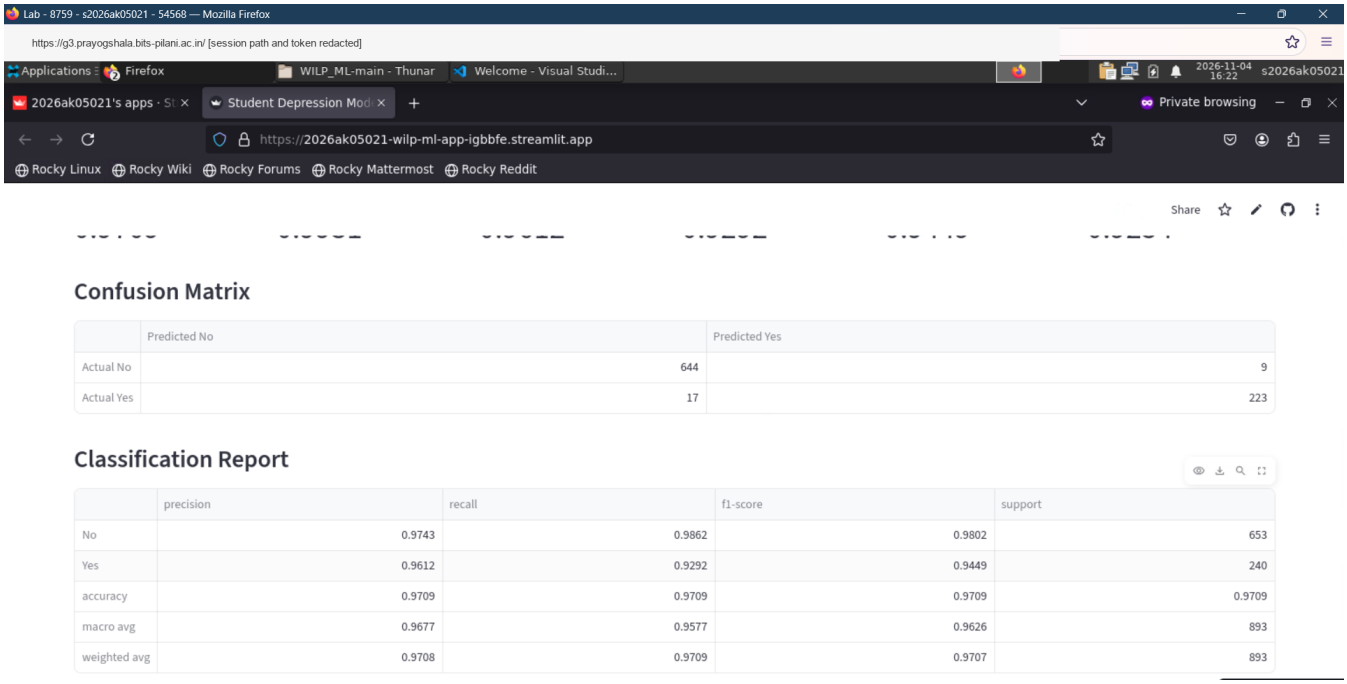
https://2026ak05021-wilp-ml-app-igbbfe.streamlit.app/

The following BITS Virtual Lab captures show all five model evaluations, with metrics, confusion matrices, and classification reports.

3. BITS Virtual Lab Evidence ? Logistic Regression

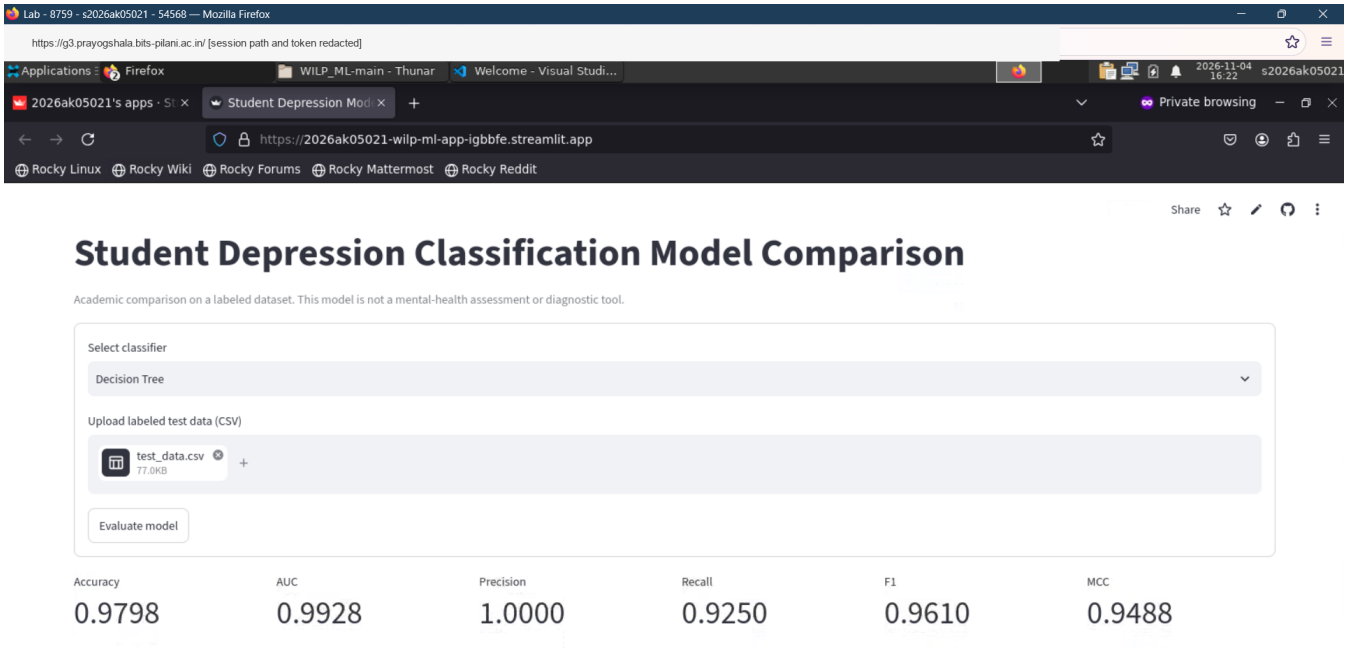


Logistic Regression: model selection, uploaded test_data.csv, and evaluation metrics. BITS lab session URL/token redacted for privacy.

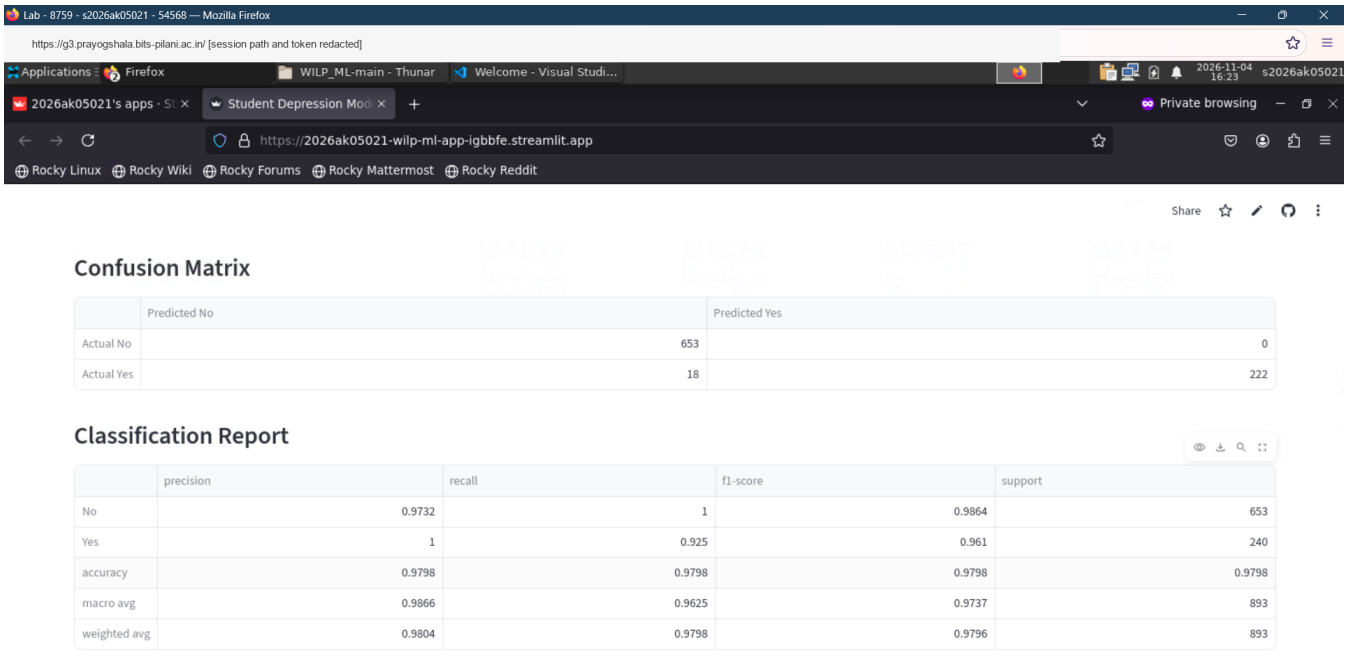


Logistic Regression: confusion matrix and classification report. BITS lab session URL/token redacted for privacy.

3. BITS Virtual Lab Evidence ? Decision Tree

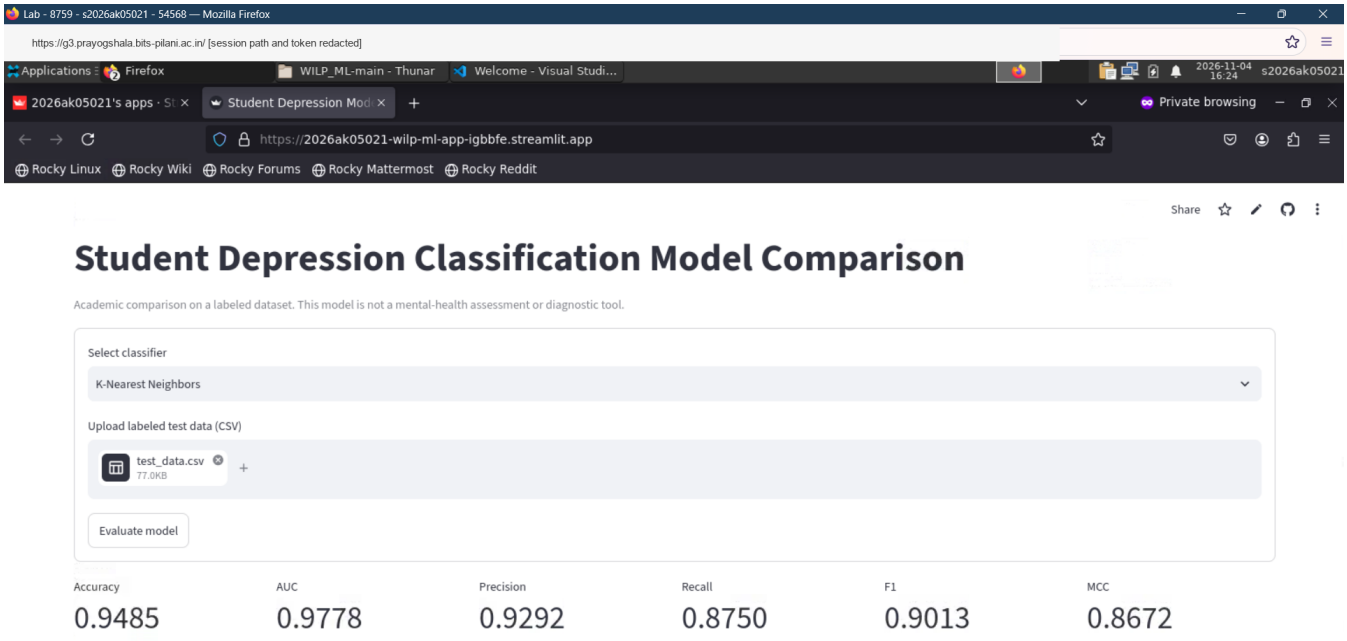


Decision Tree: model selection, uploaded test_data.csv, and evaluation metrics. BITS lab session URL/token redacted for privacy.

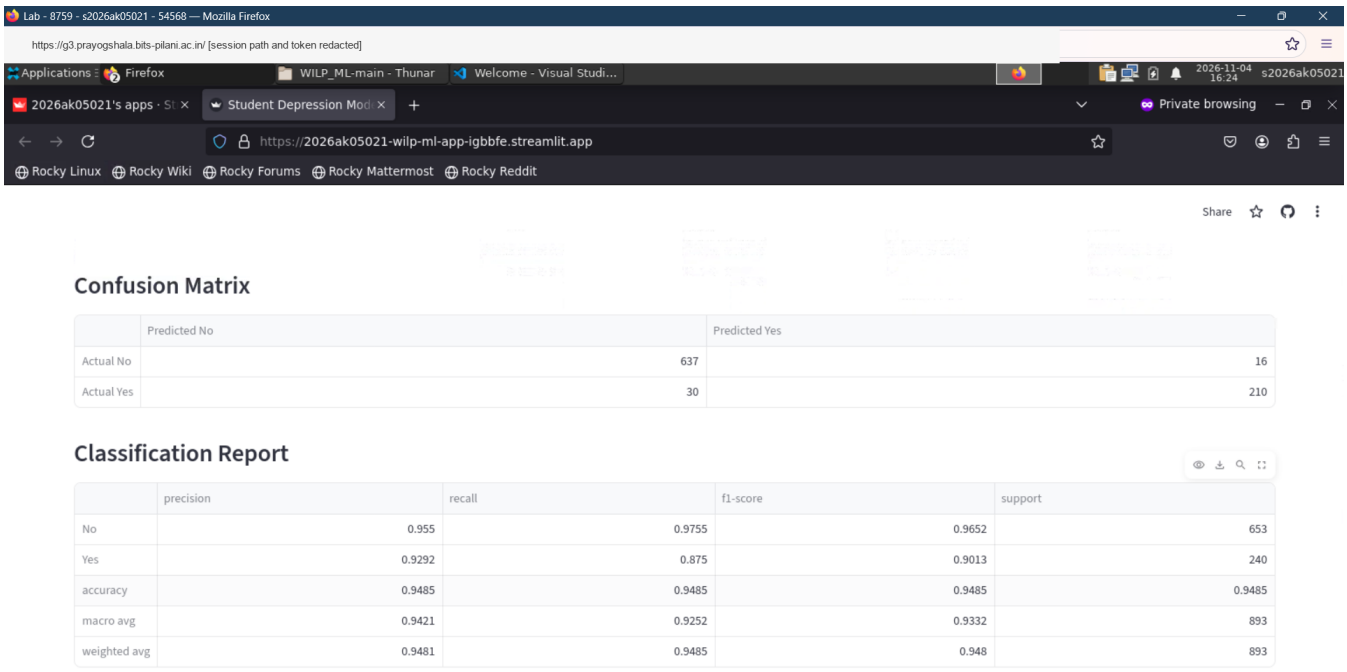


Decision Tree: confusion matrix and classification report. BITS lab session URL/token redacted for privacy.

3. BITS Virtual Lab Evidence ? K-Nearest Neighbors

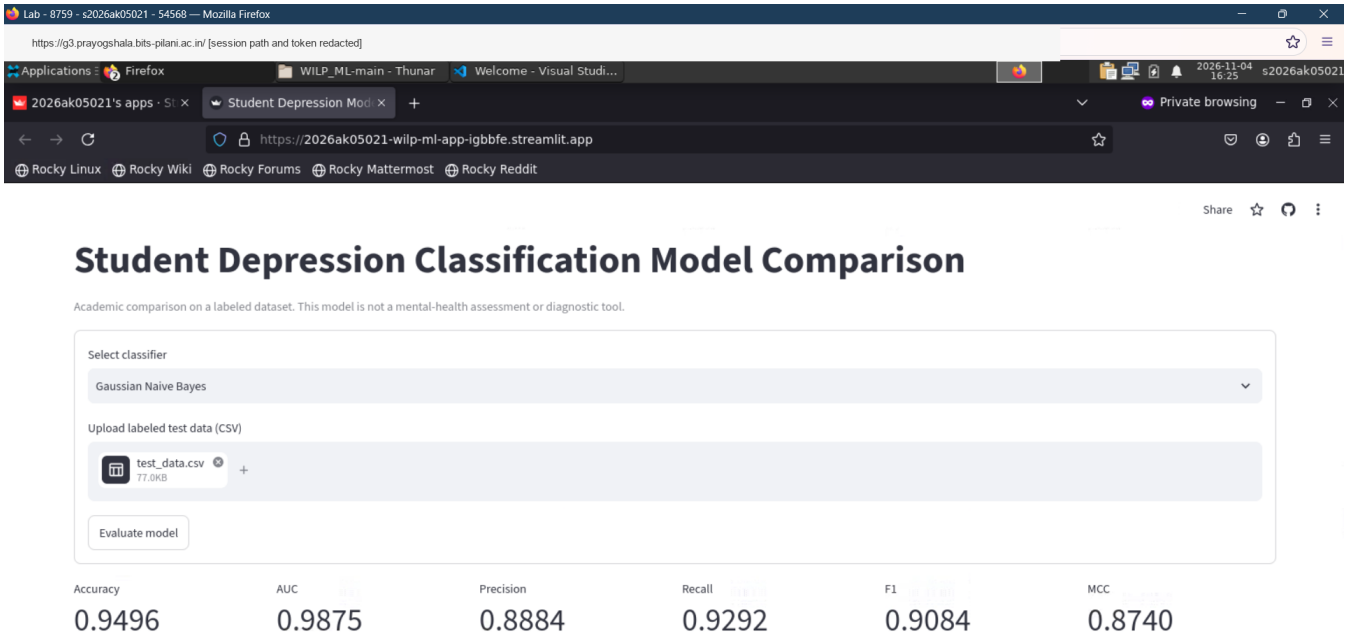


K-Nearest Neighbors: model selection, uploaded test_data.csv, and evaluation metrics. BITS lab session URL/token redacted for privacy.

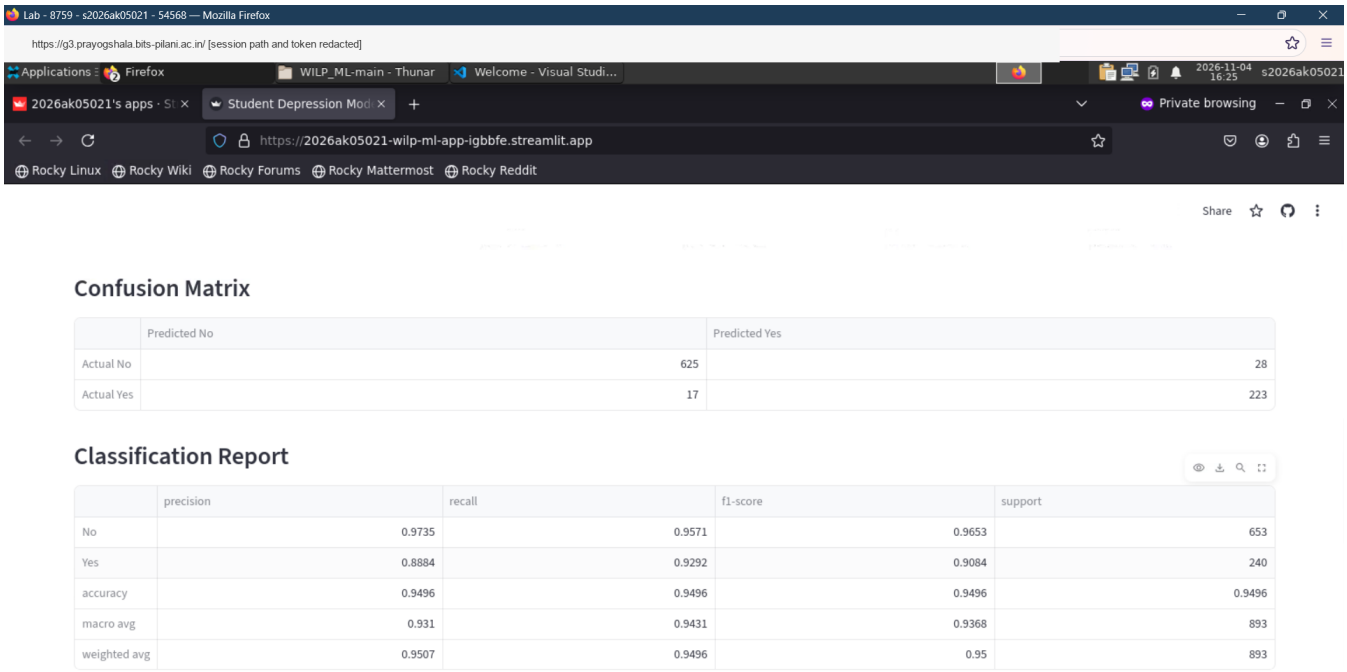


K-Nearest Neighbors: confusion matrix and classification report. BITS lab session URL/token redacted for privacy.

3. BITS Virtual Lab Evidence ? Gaussian Naive Bayes

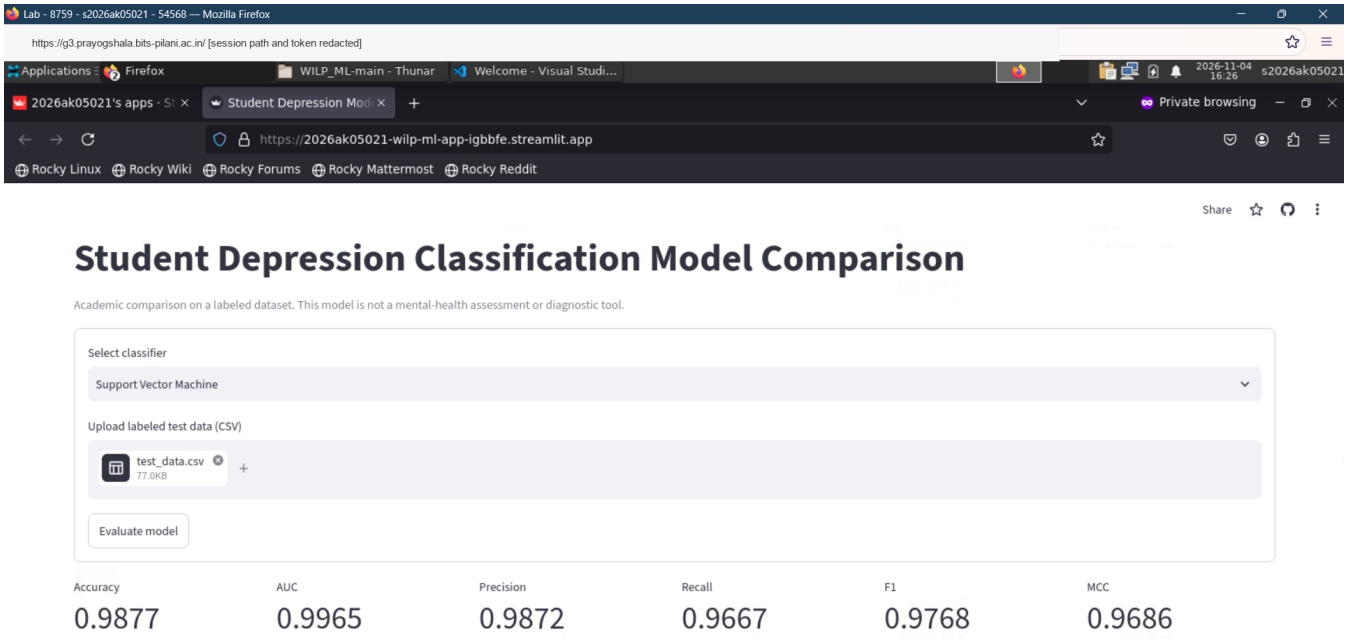


Gaussian Naive Bayes: model selection, uploaded test_data.csv, and evaluation metrics. BITS lab session URL/token redacted for privacy.

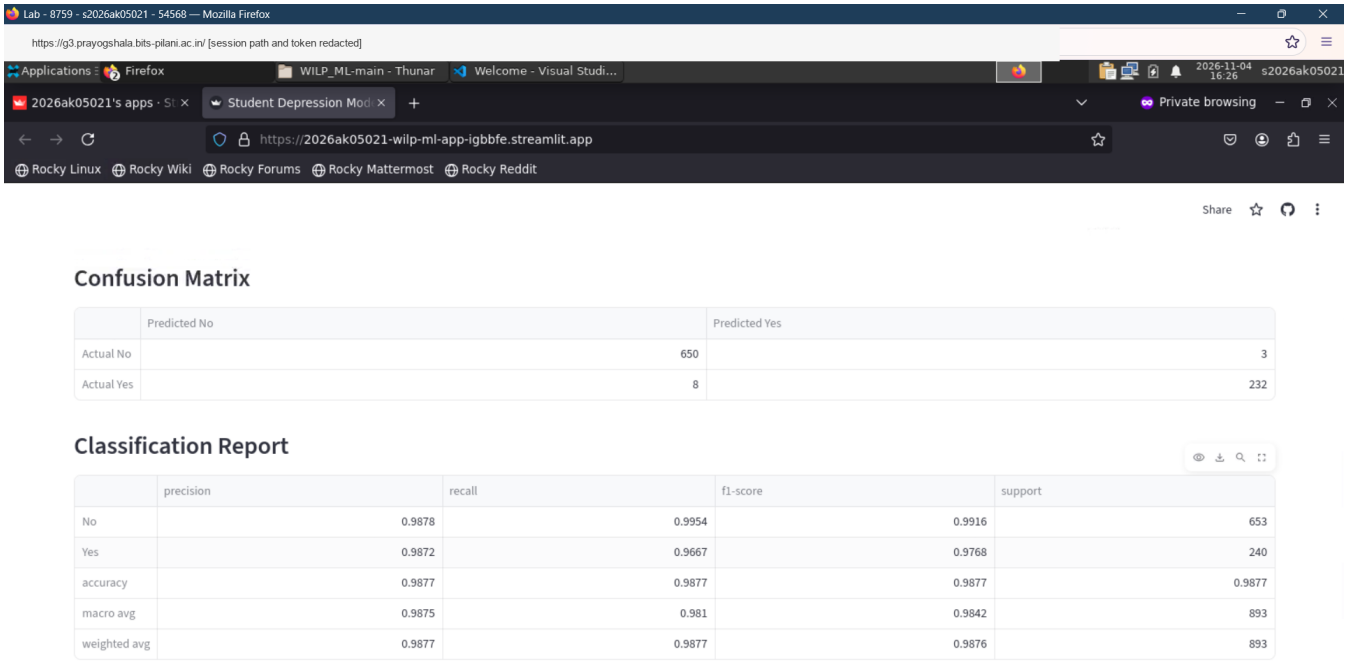


Gaussian Naive Bayes: confusion matrix and classification report. BITS lab session URL/token redacted for privacy.

3. BITS Virtual Lab Evidence ? Support Vector Machine



Support Vector Machine: model selection, uploaded test_data.csv, and evaluation metrics. BITS lab session URL/token redacted for privacy.



Support Vector Machine: confusion matrix and classification report. BITS lab session URL/token redacted for privacy.

4. Complete README.md

Student Depression Classification Model Comparison

Author: Sachin Saraswat | **Student ID:** 2026ak05021

a. Problem statement

I compared five supervised classification models using a student depression survey dataset. I built a Streamlit app where a user can upload labeled test data, select a saved model, and review its evaluation metrics. The app compares predictions with the uploaded labels; it does not retrain models or provide a clinical assessment.

b. Dataset description

The dataset is the Kaggle [Student Depression Detection Dataset](https://www.kaggle.com/datasets/mkhoirulmustaqim/student-depression-detection-dataset) by `mkhoirulmustaqim`. Kaggle describes it as survey data about demographic, academic, social, economic, and health-related factors associated with student depression. The provided CSV contains 4,464 records, 12 predictor columns, and the binary `Depression` target (`Yes`/`No`). There are no missing values in the supplied file. The CSV is used in English as provided; it was not translated or relabelled.

The predictors are `Gender`, `Age`, `Family_background`, `Relationship_status`, `Marital_status`, `Living_style`, `Current_semester`, `CGPA`, `friend_circle`, `Health_condition`, `Cost_without_study`, and `Home_division`. These meet the assignment's minimum of 12 features without adding engineered columns. The target distribution is 3,263 `No` and 1,201 `Yes`.

The dataset is used only for an academic comparison. Its associations and predictions must not be used to diagnose depression or make decisions about individual students.

c. GitHub Repository Link

- GitHub repository: 2026ak05021/WILP_ML
- Streamlit Community Cloud app: Open the app

The live Streamlit URL was verified in BITS Virtual Lab while evaluating the held-out test data.

d. Models used

The data was split once into a stratified 80% training set and 20% test set using `random_state=42` (3,571 training records and 893 test records). The same split and preprocessing pipeline were used for all five models. Numeric

predictors are median-imputed and standardized; categorical predictors are most-frequent-imputed and one-hot encoded. Preprocessing is fitted within each model pipeline using training data only.

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9709	0.9931	0.9612	0.9292	0.9449	0.9254
Decision Tree	0.9798	0.9928	1.0000	0.9250	0.9610	0.9488
K-Nearest Neighbors	0.9485	0.9778	0.9292	0.8750	0.9013	0.8672
Gaussian Naive Bayes	0.9496	0.9875	0.8884	0.9292	0.9084	0.8740
Support Vector Machine	0.9877	0.9965	0.9872	0.9667	0.9768	0.9686

Observations

ML Model Name	Observation about model performance
Logistic Regression	Provides a linear baseline and achieved 0.9709 accuracy on this held-out split.
Decision Tree	Achieved perfect precision (1.0000) with 0.9250 recall; this does not establish that it will generalize equally well to new data.
K-Nearest Neighbors	Had the lowest accuracy (0.9485) among the five on this split, while still achieving 0.8750 recall.
Gaussian Naive Bayes	Achieved 0.9496 accuracy and 0.9292 recall; its conditional-independence assumption differs from the other models.
Support Vector Machine	Achieved the highest accuracy (0.9877), AUC (0.9965), F1 (0.9768), and MCC (0.9686) on this split.
Overall winner	Support Vector Machine had the strongest results across these metrics on this single held-out split. Further validation on independent data would be needed before making broader claims.

Repository contents

- `app.py`: Streamlit CSV evaluation application
- `data_preparation.py`: target-label validation and input-column checks
- `train_models.py`: train/test split, preprocessing, model training, and metric calculation
- `requirements.txt`: pinned application dependencies
- `student_depression_dataset.csv`: Kaggle dataset used for this assignment
- `test_data.csv`: held-out test rows with all predictors and the `Depression` target
- `model/`: five saved pipelines and model/feature metadata
- `model_metrics.csv`: metrics measured on the held-out test split

Streamlit app features

The app has a labeled CSV uploader and model-selection dropdown. After the user uploads `test_data.csv` and clicks **Evaluate model**, it displays Accuracy, AUC, Precision, Recall, F1, MCC, a confusion matrix, and a classification report. It loads the selected saved pipeline and does not retrain it on uploaded data. The CSV must contain all 12 predictor columns and the `Depression` target with both `Yes` and `No` labels.

Run locally

1. Create a virtual environment with `python -m venv .venv`.
2. Activate it: on Linux/macOS, run `source .venv/bin/activate`; on Windows, run `.venv\Scripts\activate`.
3. Install dependencies with `python -m pip install -r requirements.txt`.
4. Regenerate models, metrics, and test data with `python train_models.py`.

5. Start the app with `python -m streamlit run app.py`.
6. Upload `test_data.csv`, select a model, and click **Evaluate model**.

Acknowledgment

I used AI assistance to review and revise parts of the data-preparation checks, model-training pipeline, Streamlit app, and README. I reviewed the results and will follow the course rules for acknowledging AI assistance.

Submission PDF

I included the clickable GitHub repository link, the live Streamlit app link, and screenshots captured in BITS Virtual Lab. For each of the five models, the screenshots show its evaluation metrics and its confusion matrix and classification report. The complete README is also included in the PDF.